

Why did inflation rise and fall in 2021-24?

Channels and evidence from expectations

based on Reis (2026)

Warsaw Reading Group

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How to use abstracts

What – or who – started the great depression? [☆]

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Abstract

Herbert Hoover. I develop a theory of labor market failure for the Depression based on Hoover's industrial labor program that provided industry with protection from unions in return for keeping nominal wages fixed. I find that the theory accounts for much of the depth of the Depression and for the asymmetry of the depression across sectors. The theory also can reconcile why deflation/low nominal spending apparently had such large real effects during the 1930s, but not during other periods of significant deflation.
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JEL classification: E3; I3; N1

Keywords: Great Depression; Herbert Hoover; Unionization; Deflation

Good abstract

Why did inflation rise and fall in 2021-24? Channels and evidence from expectations*

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Abstract

This article uses inflation expectations to investigate the mechanisms that linked supply and demand shocks to inflation outcomes during 2021-24. It describes several theoretical mechanisms through which shocks led to inflation, highlighting the role of expectations in this process. It uses multiple sources of expectations data for the US, EA, and UK to evaluate each of these channels. Finally, it surveys the literature that has used expectations data to make sense of the 2021-24 inflation surge. The article applies the results from this investigation to assess how well anchored inflation expectations were during the surge and at the end of it.

Wrong abstract

Motivation

Massive increase in inflation in the 2021-2023 period (after 40 years of no inflation).

The purpose of this work is to show how mechanisms through which shocks led to inflation work, and which channel they use.

This is *NOT* a paper on the underlying supply shocks!

Four theories of inflation

- Fisherian-monetarist story - inflation is the cost to deposits in banks
- New Keynesian story - inflation is the cost to workers who fight for higher wages, if rigidities exist
- Fiscal mechanism - inflation is the way to deal with debt (but works when unexpected!), especially after high deficits
- Policy choices - inflation is the cost which reduces public welfare

Fischerian-monetarist mechanism

Combining the Fischer and Taylor equations we get:

$$r_t + \bar{E}_t^m(\pi_{t+1}) = \hat{r}_t^* + \pi^* + \phi(\pi_t - \pi^*)$$

The same logic applies in the integral form

$$\pi_t = \underbrace{\pi^*}_{\text{expected inflation}} + \underbrace{\sum_{j=0}^T \phi^{-(j+1)} \bar{E}_{t,t+j-1}^m (r_{t+j} - \hat{r}_{t+j}^*)}_{\text{misjudgments of the real rate}} + \underbrace{\phi^{-T} \bar{E}_{t,T-1}^m (\pi_{t+T} - \pi^*)}_{\text{credibility of the monetary policy}}$$

Central: *markets' expectations*

Unanchoring of market expectations

Figure 2: Market expected US inflation from swaps (10-year) and indexed bonds (5y5y)



Figure: There was an unanchoring in the highest inflation period, although it is now over

Influence of inflationary shocks going forward

Table 2: The sensitivity of US long-run inflation expectations to inflation surprises

Sample	(1) 1992Q2-2019Q4	(2) 1992Q2-2019Q4	(3) 1992Q2-2019Q4	(4) 1992Q2-2019Q4
Forecast error	0.0126*** (0.0016)	0.0125*** (0.0016)	0.0118*** (0.0016)	0.0189*** (0.0019)
Observations	11,528	11,528	11,539	11,539
R-squared	0.104	0.102	0.081	0.014
Sample	2022Q1-2025Q3	2022Q1-2025Q3	2022Q1-2025Q3	2022Q1-2025Q3
Forecast error	0.0593*** (0.0056)	0.0549*** (0.0055)	0.0367*** (0.0047)	0.0528*** (0.0044)
Observations	1,314	1,314	1,316	1,316
R-squared	0.376	0.370	0.323	0.143
Quarter FE	✓	✓	✓	-
Forecaster FE	✓	✓	-	-
Horizon FE	✓	-	-	-

Figure: The realized deviations from forecasters' mistakes

New Keynesian approach

The basis of most NK models is the well known Philips curve, which involves directly expectations:

$$\pi = E^h(\pi_{t+1}) - \kappa(u - u^*) + s$$

Central: *households' expectations*

Households' expectations

Figure 4: One-year ahead household expected inflation in the US

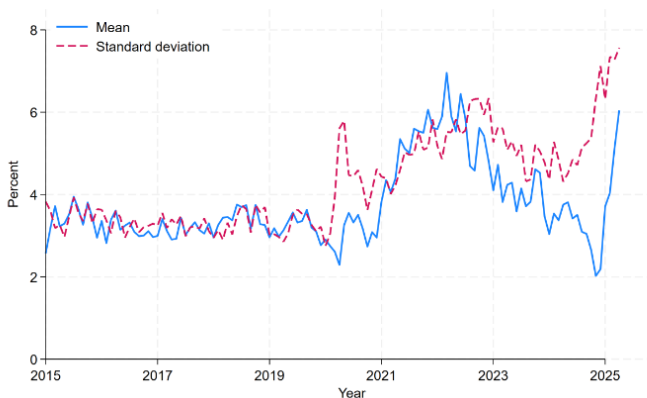


Figure: The households' expectations have unanchored as a result of the inflation surge

Not only energy and oil!

Figure 7: Energy prices do not drive inflation expectations

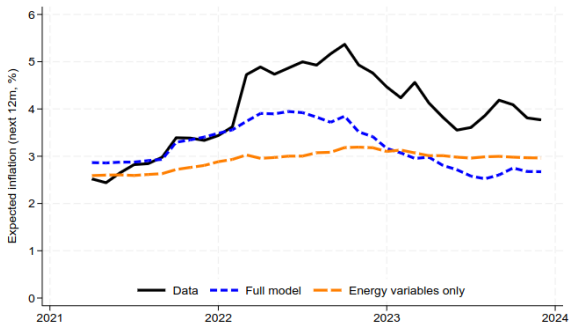


Figure: What drove households' expectations was also energy, but it explains just 25% of change

Philips curve

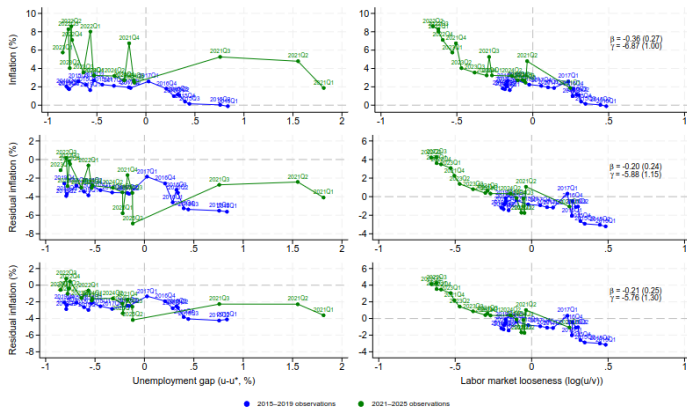


Figure: When taking into account expectations, the kink in the Philips curve remains somewhat static

Fiscal capacity

The government holding debt can "inflate it away"

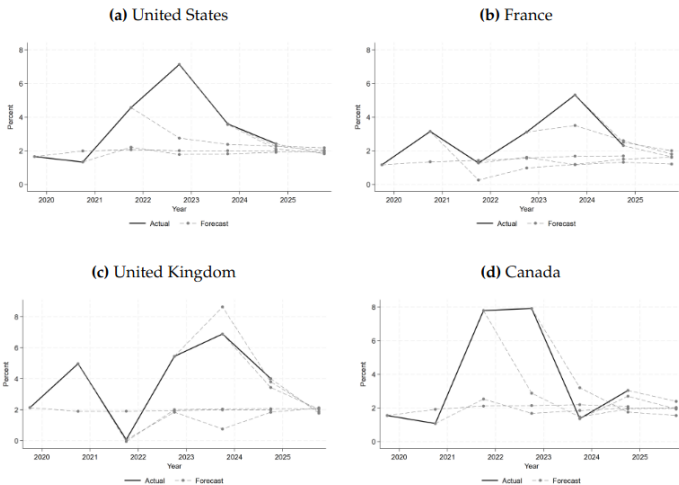
$$b_{t+1} = \frac{1 + i_t}{(1 + \pi_t)(1 + g_t)}(b_t - s_t)$$

but if bondholders expect higher inflation, they will require higher returns, and may even ask for more risk premium!

Central: *Debt institutions' expectations*

Institutions' expectations

Figure 8: IMF inflation forecasts and actual inflation in four countries



Note: This picture shows actual inflation, and the forecasts of inflation for 1 to 4 years ahead. Data comes from the April and October issues of the World Economic Outlook.

Growing importance of deficits?

Figure 9: Contributions of deficits and inflation to the unexpected increase in public debt

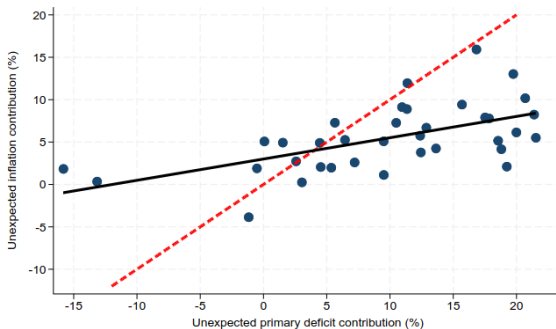


Figure: Results from Barro, Bianchi (2025) may lead to more reaction of inflation expectations to new public deficits

Policymakers' dilemma

The policymakers' expectations also influence directly the economy, for their target is:

$$\min E^P[(\pi_t - \bar{\pi})^2 + \lambda(u_t - \bar{u})^2]$$

As such, the parameter λ depends on the hawkishness of the monetary policy.

Central: *Policymakers' expectations*

Growing hawkishness

Figure 10: Expectations of inflation and unemployment gaps from FOMC members

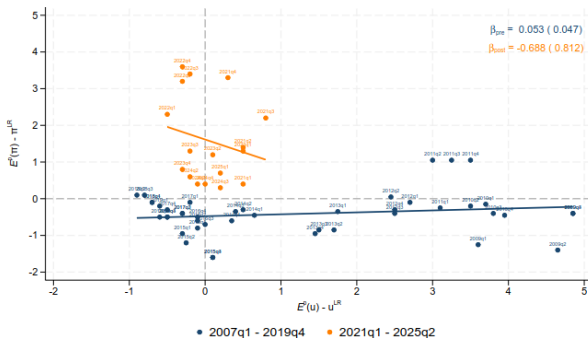


Figure: The reversal of the signs shows that hawkishness grew significantly

Takeaways

Question: Were expectations anchored?

It depends.

Question: Did expectation unanchor?

Yes, they are less anchored now than before 2020.

Question: What does it mean?

The monetary and fiscal policy is not as cost-effective as it used to be.

Shocks in the future will be more costly.

Happy Birthday!